DATABASE MANAGEMENT SYSTEM

ASSIGNMENT 9

ADVANCED SQL QUERIES USING JOINS AND SUBQUERIES

Hotel Management System

Particular Details	**Information**
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Introduction

Advanced SQL queries retrieve information from multiple related tables. Joins combine records using related columns, while subqueries use the result of one query inside another. This assignment applies INNER JOIN, LEFT JOIN, and subqueries to the Hotel Management System sample data.

1. Report on Guest Reservations Using INNER JOIN

INNER JOIN returns records with matching values in both joined tables.

```
SELECT
  r.reservation_id,
  g.guest_name,
  r.check_in_date,
  r.check_out_date,
  r.reservation_status
FROM Reservation r
INNER JOIN Guest g
  ON r.guest_id = g.guest_id
ORDER BY r.reservation_id;
```

Expected output:

reservation_id	guest_name	check_in_date	check_out_date	reservation_status
1001	Akil	2026-08-25	2026-08-27	Checked-in
1002	Rahul	2026-08-28	2026-08-30	Confirmed
1003	Priya	2026-08-29	2026-09-01	Confirmed

This query combines Reservation and Guest to show each booking with the guest's name and dates.

2. Report on Room Details for Each Reservation Using Multiple Joins

```
SELECT
  r.reservation_id,
  g.guest_name,
  rm.room_number,
  rt.type_name,
  rt.price_per_night,
  r.reservation_status
FROM Reservation r
INNER JOIN Guest g
  ON r.guest_id = g.guest_id
INNER JOIN Room rm
  ON r.room_id = rm.room_id
INNER JOIN Room_Type rt
  ON rm.room_type_id = rt.room_type_id
ORDER BY r.reservation_id;
```

Expected output:

reservation_id	guest_name	room_number	type_name	price_per_night	reservation_status
1001	Akil	205	Deluxe	4000.00	Checked-in
1002	Rahul	101	Standard	2500.00	Confirmed
1003	Priya	301	Suite	6500.00	Confirmed

The query joins four tables to show the guest, assigned room, room category, and standard nightly price.

3. Report on Room Types and Room Counts Using LEFT JOIN

LEFT JOIN includes all rows from the left table, even when there is no matching row in the right table.

```
SELECT
  rt.room_type_id,
  rt.type_name,
  COUNT(rm.room_id) AS total_rooms
FROM Room_Type rt
LEFT JOIN Room rm
  ON rt.room_type_id = rm.room_type_id
GROUP BY rt.room_type_id, rt.type_name
ORDER BY rt.room_type_id;
```

Expected output:

room_type_id	type_name	total_rooms
1	Standard	1
2	Deluxe	1
3	Suite	1

This query counts rooms in each room category. A room type with no assigned rooms would still appear with a count of zero.

4. Report on Services Priced Above the Average Using a Subquery

A subquery is a query placed inside another SQL statement. The inner query calculates the average service price.

```
SELECT service_id, service_name, service_price
FROM Service
WHERE service_price > (
  SELECT AVG(service_price)
  FROM Service
)
ORDER BY service_price DESC;
```

Expected output:

service_id	service_name	service_price
403	Airport Pickup	1200.00

The average service price is 666.67. Airport Pickup, priced at 1200.00, is the only service above the average.

5. Report on Payments Above the Average Using a Subquery

```
SELECT payment_id, reservation_id, amount, payment_status
FROM Payment
WHERE amount > (
  SELECT AVG(amount)
  FROM Payment
)
ORDER BY amount DESC;
```

Expected output:

payment_id	reservation_id	amount	payment_status
503	1003	13000.00	Pending
501	1001	9000.00	Paid

The average payment amount is 9000.00. The query returns only amounts greater than the average; the payment equal to the average is not included.

6. Report on the Highest-Priced Room Type Using a Subquery

```
SELECT room_type_id, type_name, price_per_night
FROM Room_Type
WHERE price_per_night = (
  SELECT MAX(price_per_night)
  FROM Room_Type
);
```

Expected output:

room_type_id	type_name	price_per_night
3	Suite	6500.00

The inner query finds the maximum nightly price, and the outer query returns the room type with that price.

7. Report on Services Used During Reservations

```
SELECT
  r.reservation_id,
  g.guest_name,
  s.service_name,
  su.quantity,
  s.service_price,
  (su.quantity * s.service_price) AS usage_cost
FROM Service_Usage su
INNER JOIN Reservation r
  ON su.reservation_id = r.reservation_id
INNER JOIN Guest g
  ON r.guest_id = g.guest_id
INNER JOIN Service s
  ON su.service_id = s.service_id
ORDER BY r.reservation_id, su.usage_id;
```

Expected output:

reservation_id	guest_name	service_name	quantity	service_price	usage_cost
1001	Akil	Room Service	1	500.00	500.00
1001	Akil	Laundry	2	300.00	600.00
1003	Priya	Airport Pickup	1	1200.00	1200.00

This report joins four tables and calculates the service cost as quantity multiplied by the listed service price. It is an illustrative query; taxes or other charges are not included.

Conclusion

Advanced SQL queries were implemented using INNER JOIN, LEFT JOIN, and subqueries. The reports display guest reservations, room details, room counts, services above the average price, payments above the average amount, the highest-priced room type, and service usage. These queries demonstrate how related hotel records can be combined and analysed.